



Case Study No.: 1

Title: Student Grading System with Login Authentication and Form Validation using HTML, CSS, and JavaScript

Software/Tools Required: HTML5 , CSS3 , JavaScript , Visual Studio Code , Google Chrome

Index.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Grading System</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
  <h1>Enter Subject Marks</h1>
  <form id="gradeForm">
    <div class="input-group">
      <label>JavaScript</label>
      <input
        type="number"
        id="javascript"
        placeholder="Enter Marks"
        min="0"
        max="100"
        required
      >
    </div>
    <div class="input-group">
      <label>Computer Networks</label>
      <input
        type="number"
        id="computer-networks"
        placeholder="Enter Marks"
        min="0"
        max="100"
        required
      >
    </div>
  </form>
</div>
```



```
<div class="input-group">
  <label>Compiler Construction</label>
  <input
    type="number"
    id="compiler-construction"
    placeholder="Enter Marks"
    min="0"
    max="100"
    required
  >
</div>
<div class="input-group">
  <label>Data Compression</label>
  <input
    type="number"
    id="data-compression"
    placeholder="Enter Marks"
    min="0"
    max="100"
    required
  >
</div>
<div class="input-group">
  <label>Specialization (AI / ML / Data Science)</label>
  <input
    type="number"
    id="specialization"
    placeholder="Enter Marks"
    min="0"
    max="100"
    required
  >
</div>
<button type="submit">
  Generate Result
</button>
<button type="reset" class="reset-btn">
  Reset
</button>
</form>
</div>
<script src="script.js"></script>
</body>
</html>
```



Login.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Login</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
  <h1>Student Login</h1>
  <form id="loginForm">
    <div class="input-group">
      <label>Student Name</label>
      <input
        type="text"
        id="studentName"
        placeholder="Enter Student Name"
        required
      >
    </div>
    <div class="input-group">
      <label>PRN Number</label>
      <input
        type="text"
        id="prn"
        placeholder="Enter PRN Number"
        required
      >
    </div>
    <div class="input-group">
      <label>Password</label>
      <input
        type="password"
        id="password"
        placeholder="Enter Password"
        required
      >
    </div>
    <button type="submit">
      Login
    </button>
  </form>
```



```
<p
  id="message"
  style="
    text-align:center;
    margin-top:20px;
    color:red;
    font-weight:bold;">
</p>
</div>
<script src="login.js"></script>
</body>
</html>
```

Login.js:

```
const loginForm = document.getElementById("loginForm");
loginForm.addEventListener("submit", function(event){
  event.preventDefault();
  const studentName = document.getElementById("studentName").value.trim();
  const prn = document.getElementById("prn").value.trim();
  const password = document.getElementById("password").value;
  const message = document.getElementById("message");
  message.innerHTML = "";

  if(studentName === ""){
    message.innerHTML = "Please enter Student Name.";
    return;
  }
  if(prn === ""){
    message.innerHTML = "Please enter PRN Number.";
    return;
  }

  if(!/^[0-9]+$/.test(prn)){
    message.innerHTML = "PRN should contain only digits.";
    return;
  }
  if(password.length < 8){
    message.innerHTML =
      "Password must contain at least 8 characters.";
    return;
  }
  if(!/[A-Z]/.test(password)){
    message.innerHTML =
      "Password must contain at least one uppercase letter.";
```



```
    return;
  }
  if(!/[a-z]/.test(password)){
    message.innerHTML =
      "Password must contain at least one lowercase letter.";
    return;
  }
  if(!/[0-9]/.test(password)){
    message.innerHTML =
      "Password must contain at least one number.";
    return;
  }
  if(!/[!@#%$%^&*(),.?":{}|<>]/.test(password)){
    message.innerHTML =
      "Password must contain at least one special character.";
    return;
  }
  const student = {
    name: studentName,
    prn: prn
  };
  localStorage.setItem(
    "loggedInStudent",
    JSON.stringify(student)
  );
  window.location.href = "index.html";
});
```

Result.html:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport"
    content="width=device-width, initial-scale=1.0">
  <title>Student Result</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container result-container">
  <h1>Student Grade Report</h1>
  <div id="resultCard">
    <!-- Result will be loaded here -->
  </div>
```



```
<div class="button-group">
  <button onclick="goBack()">
    Back
  </button>
</div>
</div>
<script src="result.js"></script>
</body>
</html>
```

Result.js:

```
const student = JSON.parse(localStorage.getItem("studentResult"));
const resultCard = document.getElementById("resultCard");
if (student === null) {
  resultCard.innerHTML = `
    <h2 style="color:red;text-align:center;">
      No Student Data Found
    </h2>
    <p style="text-align:center;">
      Please enter student details first.
    </p>
  `;
}
else {
  resultCard.innerHTML = `
    <table>
      <tr>
        <th colspan="2">
          Student Information
        </th>
      </tr>
      <tr>
        <td><strong>Student Name</strong></td>
        <td>${student.studentName}</td>
      </tr>
      <tr>
        <td><strong>Roll Number</strong></td>
        <td>${student.rollNumber}</td>
      </tr>
    </table>
    <br>
    <table>
      <tr>
        <th>Subject</th>
```



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Experiment-2

```
<th>Marks</th>
</tr>
<tr>
  <td>JavaScript</td>
  <td>${student.javascript}</td>
</tr>
<tr>
  <td>Computer Networks</td>
  <td>${student.computerNetworks}</td>
</tr>
<tr>
  <td>Compiler Construction</td>
  <td>${student.compilerConstruction}</td>
</tr>
<tr>
  <td>Data Compression</td>
  <td>${student.dataCompression}</td>
</tr>
<tr>
  <td>Specialization</td>
  <td>${student.specialization}</td>
</tr>
</table>
<br>
<table>
  <tr>
    <th colspan="2">
      Result Summary
    </th>
  </tr>
  <tr>
    <td><strong>Total Marks</strong></td>
    <td>${student.total} / 500</td>
  </tr>
  <tr>
    <td><strong>Percentage</strong></td>
    <td>${student.percentage}%</td>
  </tr>
  <tr>
    <td><strong>Grade</strong></td>
    <td class="grade">${student.grade}</td>
  </tr>
  <tr>
    <td><strong>Status</strong></td>
```



```
<td class="${student.status === "Pass" ? "success" : "fail"}">
    ${student.status}
</td>
</tr>
<tr>
    <td><strong>Remarks</strong></td>
    <td>${student.remarks}</td>
</tr>
</table>
`;
}
function goBack() {
    window.location.href = "index.html";
}
```

Script.js:

```
const loginStudent = JSON.parse(localStorage.getItem("loggedInStudent"));
if (!loginStudent) {
    window.location.href = "login.html";
}
const gradeForm = document.getElementById("gradeForm");
gradeForm.addEventListener("submit", function (event) {
    event.preventDefault();
    const studentName = loginStudent.name;
    const rollNumber = loginStudent.prn;
    const javascript = Number(document.getElementById("javascript").value);
    const computerNetworks = Number(document.getElementById("computer-networks").value);
    const compilerConstruction = Number(document.getElementById("compiler-construction").value);
    const dataCompression = Number(document.getElementById("data-compression").value);
    const specialization = Number(document.getElementById("specialization").value);
    const marks = [
        javascript,
        computerNetworks,
        compilerConstruction,
        dataCompression,
        specialization
    ];
    const subjectNames = [
        "JavaScript",
        "Computer Networks",
        "Compiler Construction",
        "Data Compression",
    ]
```




```
"Specialization"
];
for (let i = 0; i < marks.length; i++) {
  if (isNaN(marks[i])) {
    alert("Please enter " + subjectNames[i] + " marks.");
    return;
  }
  if (marks[i] < 0 || marks[i] > 100) {
    alert(subjectNames[i] + " marks should be between 0 and 100.");
    return;
  }
}
let total = 0;
for (let mark of marks) {
  total += mark;
}
const percentage = (total / 500) * 100;
let status = "Pass";
for (let mark of marks) {
  if (mark < 35) {
    status = "Fail";
    break;
  }
}
let grade = "";
if (status === "Fail") {
  grade = "F";
}
else if (percentage >= 90) {
  grade = "A+";
}
else if (percentage >= 80) {
  grade = "A";
}
else if (percentage >= 70) {
  grade = "B";
}
else if (percentage >= 60) {
  grade = "C";
}
else if (percentage >= 50) {
  grade = "D";
}
else {
```



```
    grade = "F";
  }
  let remarks = "";
  switch (grade) {
    case "A+":
      remarks = "Outstanding";
      break;
    case "A":
      remarks = "Excellent";
      break;
    case "B":
      remarks = "Very Good";
      break;
    case "C":
      remarks = "Good";
      break;
    case "D":
      remarks = "Needs Improvement";
      break;
    default:
      remarks = "Better Luck Next Time";
  }
  const studentResult = {
    studentName,
    rollNumber,
    javascript,
    computerNetworks,
    compilerConstruction,
    dataCompression,
    specialization,
    total,
    percentage: percentage.toFixed(2),
    grade,
    status,
    remarks
  };
  localStorage.setItem(
    "studentResult",
    JSON.stringify(studentResult)
  );
  window.location.href = "result.html";
});
```



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Experiment-2

Style.css:

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family: Ballance;
}

body{
  background:linear-gradient(135deg,#99c6ee,#a4e5f0);
  display:flex;
  justify-content:center;
  align-items:center;
  min-height:100vh;
}

.container{
  width: 1000px;
  background:#c7e0f4;
  padding:30px;
  border-radius:15px;
  box-shadow:0 10px 25px rgba(0,0,0,.2);
}

h1{
  text-align:center;
  margin-bottom:25px;
  color:#0e0e0f;
}

.input-group{
  margin-bottom:15px;
}

label{
  display:block;
  margin-bottom:6px;
  font-weight:bold;
}

input{
  width:100%;
  padding:10px;
  border:1px solid #d2e4fd;
```



```
border-radius:8px;
font-size:15px;
}

input:focus{
  outline:none;
  border-color:#0d6efd;
}

button{
  width:100%;
  padding:12px;
  background:#0c48a1;
  color:white;
  border:none;
  border-radius:8px;
  font-size:16px;
  cursor:pointer;
}

button:hover{
  background:#084298;
}

#result{
  margin-top:20px;
  display:none;
  padding:15px;
  background:#f5f8ff;
  border-radius:10px;
}

table{
  width:100%;
  border-collapse:collapse;
}

td{
  padding:10px;
  border-bottom:1px solid #ddd;
}

.success{
  color:green;
```



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Experiment-2

```
font-weight:bold;
}
.fail{
color:red;
font-weight:bold;
}
```

Output:

The screenshot shows a web browser window with the URL `C:/Users/hp/Documents/24070521148_ShrutiBawankar_C1/JavaScript_Lab/Case%20Study/CaseStudy-3/login.html`. The page displays a "Student Login" form with three input fields: "Student Name" (containing "Enter Student Name"), "PRN Number" (containing "24070521148"), and "Password" (containing "Enter Password"). A blue "Login" button is at the bottom. A red error message "Please fill out this field." is displayed next to the PRN Number field.

The screenshot shows the same "Student Login" form, but now the "Student Name" field contains "Shruti Bawankar" and the "Password" field contains "*****". The "PRN Number" field still contains "24070521148". A red error message "Password must contain at least one special character." is displayed below the "Login" button.



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Experiment-2

The screenshot displays two pages of a web application. The top page, 'index.html', is titled 'Enter Subject Marks' and contains a form for entering marks for various subjects. The bottom page, 'result.html', displays the student's information and a table of their marks, along with a result summary.

Enter Subject Marks

JavaScript: 89

Computer Networks: 87

Compiler Construction: 90

Data Compression: 93

Specialization (AI / ML / Data Science): 95

Generate Result

Reset

Student Information

Student Name: Shruti Bawankar

Roll Number: 24070521148

Subject	Marks
JavaScript	89
Computer Networks	87
Compiler Construction	90
Data Compression	93
Specialization	95

Result Summary

Total Marks: 454 / 500

Percentage: 90.80%

Grade: A+

Status: Pass

Remarks: Outstanding

Back

Result: The Student Grading System was successfully developed to validate user credentials, process student marks, calculate grades, and display the results accurately on a separate result page.

Conclusion: This project successfully demonstrates the implementation of JavaScript form validation and control structures, providing a simple, secure, and user-friendly web application for student result management.